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# Rapport d'Enseignement d'Intégration ST5

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## Contrôle de la pollution acoustique intérieure

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# 1 Introduction

The acoustics of concert halls have evolved over the decades, with fundamental research in the 1960s helping to understand noise sources and optimize the use of acoustic liners in enclosed spaces [3]. These early studies laid the foundations for current methods used in noise reduction in concert settings.

In the 1980s, research introduced more efficient sound-absorbing materials, such as multilayer panels and perforated foams, which helped better control reverberation in concert halls [4]. The advent of numerical simulations also allowed for more precise modeling of sound propagation. The 1990s saw the integration of honeycomb-structured materials, improving sound absorption across a broader frequency range [5]. These materials were combined with active acoustic systems, allowing for real-time adjustment of reverberation based on the specific needs of the performance type.

In the 2000s, techniques such as genetic algorithms emerged, optimizing the configuration of acoustic elements in halls, particularly regarding passive and active liners [6]. Advances in metamaterials allowed for the introduction of adaptive solutions that could respond to real-time frequency variations [7]. Today, modern technologies such as artificial intelligence and smart materials enable dynamic optimization of acoustic environments, providing optimal acoustic comfort for an increasingly diverse range of performances.

Despite these advances, the challenge of effectively absorbing unwanted noise in concert halls while preserving their architectural integrity and aesthetic design remains a major concern. Excessive reverberation or poorly distributed reflections can deteriorate the clarity, warmth, and intimacy of musical performances, particularly in rectangular halls where reflections from the rear wall may cause acoustic distortions. Designing walls capable of absorbing targeted frequencies efficiently, without compromising visual harmony or incurring high material costs, thus represents a key problem in modern acoustic engineering.

In this context, our project focuses on the development and numerical simulation of an optimized absorbent wall located at the back of an idealized rectangular concert hall. Through advanced modeling of the Helmholtz equation with mixed boundary conditions, we aim to evaluate and optimize the acoustic performance of such walls. This research not only serves as a proof of concept for a new design approach but also lays the foundation for our startup initiative, **AcouFun**, which applies mathematical modeling and numerical optimization to the design of efficient, ecologic, and cost-effective acoustic solutions.

## 2 The birth of AcouFun

AcouFun was founded from the collaboration of engineers and researchers passionate about mathematical modeling, orchestras, and architectural design. The project originated from the observation that many concert halls, even modern ones, struggle to achieve an ideal balance between sound absorption, ecologic design, and economic feasibility. Our goal is to bridge this gap by developing customized acoustic walls that combine scientific precision with artistic sensibility.

The main objectives of AcouFun are :

1. To design optimal sound-absorbing walls for concert halls, capable of minimizing unwanted reverberations and reflections, particularly those coming from the rear boundaries of rectangular spaces.
2. To offer cost-effective and non-invasive acoustic solutions, maintaining the architectural and aesthetic coherence of the performance space.
3. To integrate advanced mathematical and computational methods, such as PDE-based optimization and numerical simulation, into the acoustic design process.
4. To tailor each project according to the budget and specific acoustic needs of our clients, adapting the distribution, thickness, and material properties of the absorbent structures accordingly.

A distinctive feature of AcouFun is its scientific approach as a seal of quality. Our methodology combines careful material selection—balancing acoustic absorption, ecological criteria and cost—with identification of target frequencies based on the concert arrangement, ensuring precise control of the acoustic response.

This multidisciplinary approach allows AcouFun to provide personalized, scientifically validated acoustic solutions, making high-quality sound environments accessible to a wide range of concert halls and cultural venues.



FIGURE 1 – AcouFun

### 3 Model Considered

We model the complex acoustic pressure  $u$  satisfying the Helmholtz equation in  $\Omega$ , with mixed boundary conditions : non-homogeneous Dirichlet for the incoming noise, Neumann for the lateral walls, and a homogeneous Robin condition on the control boundary  $\Gamma_R$  representing the noise absorption wall :

$$\begin{cases} (\Delta + k^2)u = 0 & \text{in } \Omega, \\ u_{\text{phys}} = g & \text{on } \Gamma_D, \\ \partial_\nu u = 0 & \text{on } \Gamma_N, \\ \partial_\nu u + \chi\alpha \text{Tr } u = 0 & \text{on } \Gamma_R, \end{cases} \quad (1)$$

where  $g$  represents the noise generated by the concert, coming from the upper boundary of  $\Omega$ , as illustrated for  $\Gamma_1$  in Figure 5. Here,  $\chi$  is a characteristic function of the regions filled with porous material, defined by Equation (2), whose distribution we seek to optimize.

$$\forall x \in \partial\Omega, \quad \chi(x) = \begin{cases} 1, & \text{if porous material is present at } x, \\ 0, & \text{if no porous material is present at } x. \end{cases} \quad (2)$$

To model absorption, the Robin boundary condition must involve a complex impedance. For the wall to absorb acoustic energy and to ensure wellposedness, one requires

$$\text{Re}(\alpha(x)) > 0.$$

The objective of the optimization is to minimize the interior acoustic energy, which is

$$J(\chi) = \frac{1}{2} \int_{\Omega} |u(\chi)|^2 dx \quad (3)$$

and the admissible set considered is

$$U_{ad}(\beta) = \left\{ \chi \in L^\infty(\Gamma) : \chi(x) \in \{0, 1\} \text{ on } \partial\Omega, \quad \int_{\partial\Omega} \chi d\mu = \beta \right\},$$

where  $\beta$  denotes the amount of absorbent material available. At all times,  $\beta$  is bounded by  $\int_{\Gamma_R} 1, d\mu$ . However, for the purposes of our startup project, we will instead define or constrain  $\beta$  in terms of its total cost. For simplicity, we consider  $\beta \in (0, 1)$  as a percentage.

Further considerations on the selection of this mathematical model are provided in the following section, Theoretical Study, which also explains the computation of the energy derivative, crucial for performing our optimization.

### 3.1 Modeling of the source $g$

For simplification reasons, we considered that the concert hall will host symphonic orchestra. These orchestra have a particular geometry that enables us to have more information on the frequencies delivered.



FIGURE 2 – Geometry of the orchestra

We modeled the orchestra as three distinct sections : the violins and violas on the left, the wind instruments and percussion in the middle, and the cello and double bass on the right. Each instrument produces sound within a frequency range contained in  $[20 \text{ Hz}, 4000 \text{ Hz}]$ .

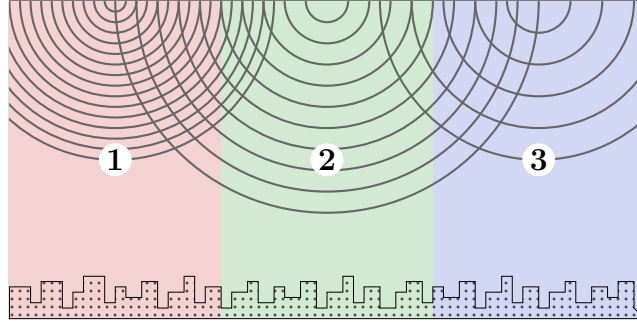


FIGURE 3 – Model of the noise sources divided into frequency sectors according to the typical distribution of instruments in an orchestra.

We suppose that each source produces a sound composed of up to 30 notes as there are 30 instruments in each section. We consider that the sound is pure and therefore there are no harmonics. For one note, the signal is :

$$s(t) = A \sin(2\pi f_0 t) \quad (4)$$

where  $A$  is the amplitude and  $f_0$  the fundamental frequency of the note. Therefore the source can be modelled as :

$$s(t) = \sum_{i=1}^{30} A_i \sin(2\pi f_i t) \quad (5)$$

If we add the harmonics, we will suppose that the amplitude of the harmonics decrease exponentially. It leads us to this formula :

$$s(t) = \sum_{i=1}^{30} A_i \sum_{n=1}^K \frac{1}{n} \sin(2\pi n f_i t) \quad (6)$$

where  $K$  is such that  $\frac{f_{max}}{f_i} - 1 < K \leq \frac{f_{max}}{f_i}$  with  $f_{max} = 4000Hz$ .

$$\begin{cases} s_1(t) = \sum_{i=1}^{30} A_i \sum_{n=1}^K \frac{1}{n} \sin(2\pi n f_{i,1} t), & \forall i \in \llbracket 1; 30 \rrbracket, f_{i,1} \in [125, 1250] Hz \\ s_2(t) = \sum_{i=1}^{30} A_i \sum_{n=1}^K \frac{1}{n} \sin(2\pi n f_{i,2} t), & \forall i \in \llbracket 1; 30 \rrbracket, f_{i,2} \in [30, 4000] Hz \\ s_3(t) = \sum_{i=1}^{30} A_i \sum_{n=1}^K \frac{1}{n} \sin(2\pi n f_{i,3} t), & \forall i \in \llbracket 1; 30 \rrbracket, f_{i,3} \in [30, 400] Hz \end{cases} \quad (7)$$

The frequency ranges were chosen according to Figure 4 [1].

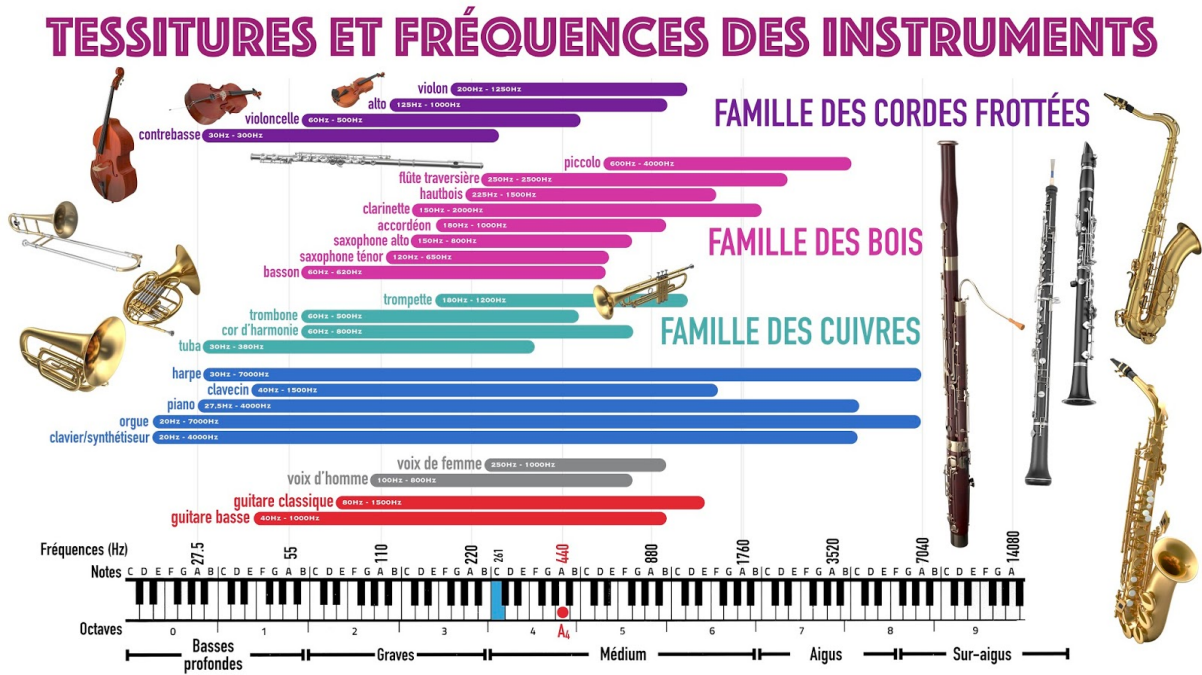


FIGURE 4 – Frequency range of each instrument

### 3.2 Theoretical Study

We consider that both side boundaries are totally reflective, hence subject to a Neumann boundary condition. As stated previously, the Helmholtz equation for this model is the following :

$$\begin{cases} (\Delta + k^2)u = 0 & \text{in } \Omega, \\ \partial_\nu u = 0 & \text{on } \Gamma_N, \\ u = g & \text{on } \Gamma_D, \\ \partial_\nu u + \chi\alpha u = 0 & \text{on } \Gamma_R, \end{cases} \quad (8)$$

where  $k > 0$  is a fixed wavenumber and  $g$  is a prescribed source representing the noise generated by the concert, coming from the upper boundary of  $\Omega$ .

Additionally, we consider instead the scenario where both side boundaries are open-air, and therefore subject to a Sommerfeld-type boundary condition. We have

$$\begin{cases} (\Delta + k^2)u = 0 & \text{on } \Omega \\ \partial_\nu u - iku = 0 & \text{on } \Gamma_S \\ u = g & \text{on } \Gamma_D \\ \partial_\nu u + \chi\alpha u = 0 & \text{on } \Gamma_R \end{cases} \quad (9)$$

We will compute the derivatives of the energies associated with both Problem 8 and Problem 9, and compare them to provide some insights on the selection of the model and on how to implement our numerical optimization problem. Diagrams on Figure 5 show the set up.

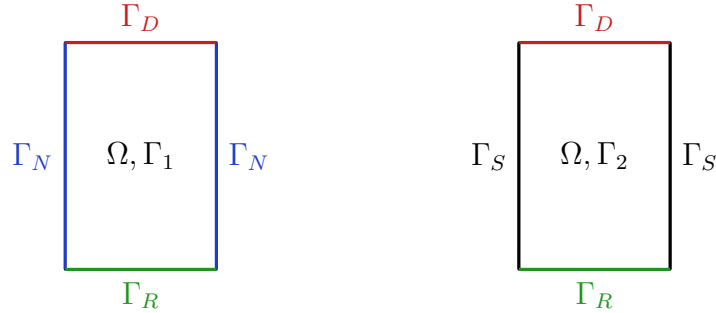


FIGURE 5 – Partitions  $\Gamma_1$ ,  $\Gamma_2$  of the boundary of  $\Omega$ .

Let the space

$$V(\Omega) := \{v \in H^1(\Omega) : \text{Tr } v = 0 \text{ on } \Gamma_D\},$$

endowed with the norm

$$\|u\|_{V(\Omega)}^2 := \int_{\Omega} |\nabla u|^2 dx,$$

which is equivalent to  $\|\cdot\|_{H^1(\Omega)}$  due to the Poincaré inequality, valid as long as  $\Gamma_D$  has nonzero measure. From the continuity of the trace operator it follows directly that  $V$  is a closed subspace of  $H^1(\Omega)$ , thus a Hilbert space.

Let  $\tilde{g}$  be the lifting function associated to the common Dirichlet condition, thus solving Problem 10.

$$\begin{cases} \Delta \tilde{g} = 0 \\ \partial_\nu \tilde{g} = 0 & \text{on } \Gamma \setminus \Gamma_D \\ \text{Tr } \tilde{g} = g & \text{on } \Gamma_D. \end{cases} \quad (10)$$



We claim that  $\tilde{g}$  is well defined. Indeed, the weak formulation of Problem 10 is

$$\langle \phi_0, v \rangle_{V(\Omega)} = -\langle \nabla \phi, \nabla v \rangle_{L^2(\Omega)} \quad \forall v \in V(\Omega) \quad (11)$$

where  $\tilde{g} = \phi + \phi_0$ , and  $\phi$  is a function in  $H^1(\Omega)$  such that  $\text{Tr}|_{\Gamma_D} \phi = g$ . By Riesz theorem this problem is well posed.

Solutions to Problem (8) and Problem (9) belong to  $V$ , then the variational formulations are

$$\begin{aligned} FV_r(\chi, \tilde{u}_R^r, \tilde{u}_I^r, v_R, v_I) = & \int_{\Omega} (\nabla \tilde{u}_I^r \nabla v_I - \nabla \tilde{u}_R^r \nabla v_R) dx + k^2 \int_{\Omega} (\tilde{u}_R^r v_R - \tilde{u}_I^r v_I + \tilde{g}_R v_R - \tilde{g}_I v_I) dx \\ & + \int_{\Gamma} \chi [(\alpha_I \tilde{u}_R^r + \alpha_R \tilde{u}_I^r) v_I - (\alpha_R \tilde{u}_R^r - \alpha_I \tilde{u}_I^r) v_R] d\mu \\ & + \int_{\Gamma_R} \chi [(\alpha_I \tilde{g}_R + \alpha_R \tilde{g}_I) v_I - (\alpha_R \tilde{g}_R - \alpha_I \tilde{g}_I) v_R] d\mu = 0 \quad \forall (v_R, v_I) \in V(\Omega) \times V(\Omega) \end{aligned} \quad (12)$$

for the reflective case and

$$\begin{aligned} FV_o(\chi, \tilde{u}_R^o, \tilde{u}_I^o, v_R, v_I) = & \int_{\Omega} (\nabla \tilde{u}_I^o \nabla v_I - \nabla \tilde{u}_R^o \nabla v_R) dx + k^2 \int_{\Omega} (\tilde{u}_R^o v_R - \tilde{u}_I^o v_I + \tilde{g}_R v_R - \tilde{g}_I v_I) dx \\ & + k \int_{\Gamma_S} (\tilde{u}_R^o v_I + \tilde{u}_I^o v_R + \tilde{g}_R v_I + \tilde{g}_I v_R) d\mu \\ & + \int_{\Gamma_R} \chi [(\alpha_I \tilde{u}_R^o + \alpha_R \tilde{u}_I^o) v_I - (\alpha_R \tilde{u}_R^o - \alpha_I \tilde{u}_I^o) v_R] d\mu \\ & + \int_{\Gamma_R} \chi [(\alpha_I \tilde{g}_R + \alpha_R \tilde{g}_I) v_I - (\alpha_R \tilde{g}_R - \alpha_I \tilde{g}_I) v_R] d\mu = 0 \quad \forall (v_R, v_I) \in V(\Omega) \times V(\Omega) \end{aligned} \quad (13)$$

for the open air boundary condition, where  $\tilde{u}^i = u^i - \tilde{g}$  for  $i \in r, o$ , meaning *reflexive* and *open air* respectively, and  $R, I$  denote the real and imaginary parts of the functions. The given variational formulations correspond to the subtraction between the variational formulations of the real and imaginary parts of both problems.

Let  $\mathcal{J}_r, \mathcal{J}_o$  and  $L_r, L_o$  be the energies and Lagrangians associated to Problem 8 and Problem 9 respectively, then

$$\begin{aligned} J_i(\omega_R, \omega_I) &= \int_{\Omega} [(\omega_R + \tilde{g}_R)^2 + (\omega_I + \tilde{g}_I)^2] dx \quad \forall \omega_R, \omega_I \in V(\Omega) \\ \mathcal{L}_i(\chi, \omega_R, \omega_I, q_R, q_I) &= FV_i(\chi, \omega_R, \omega_I, q_R, q_I) + J_i(\omega_R, \omega_I) \quad \forall \chi \in L^\infty(\Gamma), \omega_R, \omega_I, q_R, q_I \in V(\Omega) \end{aligned} \quad (14)$$

for  $i \in \{r, o\}$ . The Fréchet derivatives of the energies are given by

$$\langle J'_i(\chi), \chi_0 \rangle = \left\langle \frac{\partial \mathcal{L}_i}{\partial \chi}(\chi, \tilde{u}_R^i(\chi), \tilde{u}_I^i(\chi), p_R^i(\chi), p_I^i(\chi)), \chi_0 \right\rangle, \quad (15)$$

for  $i \in \{r, o\}$  where  $p^i$  is the solution to the respective version of the adjoint Problem 16.

$$\begin{cases} \left\langle \frac{\partial \mathcal{L}_i}{\partial \omega_R}(\chi, \tilde{u}_R^i(\chi), \tilde{u}_I^i(\chi), p_R^i(\chi), p_I^i(\chi)), v_R \right\rangle = 0 \\ \left\langle \frac{\partial \mathcal{L}_i}{\partial \omega_I}(\chi, \tilde{u}_R^i(\chi), \tilde{u}_I^i(\chi), p_R^i(\chi), p_I^i(\chi)), v_I \right\rangle = 0, \end{cases} \quad \forall (v_R, v_I) \in V(\Omega) \times V(\Omega). \quad (16)$$

This gives

$$\begin{aligned}
& \int_{\Omega} (-\nabla p_R^r \nabla v_R + k^2 p_R^r v_R) dx + \int_{\Gamma_R} \chi (\alpha_I p_I^r v_R - \alpha_R p_R^r v_R) d\mu \\
& \quad + \int_{\Omega} 2(\tilde{u}_R^r + \tilde{g}_R) v_R dx = 0 \quad \forall v_R \in V(\Omega) \\
& \int_{\Omega} (\nabla p_I^r \nabla v_I - k^2 p_I^r v_I) dx + \int_{\Gamma_R} \chi (\alpha_R p_I^r v_I + \alpha_I p_R^r v_I) d\mu \\
& \quad + \int_{\Omega} 2(\tilde{u}_I^r + \tilde{g}_I) v_I dx = 0 \quad \forall v_I \in V(\Omega),
\end{aligned} \tag{17}$$

for the reflective case and

$$\begin{aligned}
& \int_{\Omega} (-\nabla p_R^o \nabla v_R + k^2 p_R^o v_R) dx + \int_{\Gamma_R} \chi (\alpha_I p_I^o v_R - \alpha_R p_R^o v_R) d\mu \\
& \quad + \int_{\Gamma_S} k p_I^o v_R d\mu + \int_{\Omega} 2(\tilde{u}_R^o + \tilde{g}_R) v_R dx = 0 \quad \forall v_R \in V(\Omega) \\
& \int_{\Omega} (\nabla p_I^o \nabla v_I - k^2 p_I^o v_I) dx + \int_{\Gamma_R} \chi (\alpha_R p_I^o v_I + \alpha_I p_R^o v_I) d\mu \\
& \quad + \int_{\Gamma_S} k p_R^o v_I d\mu + \int_{\Omega} 2(\tilde{u}_I^o + \tilde{g}_I) v_I dx = 0 \quad \forall v_I \in V(\Omega),
\end{aligned} \tag{18}$$

for the open air boundary condition.

We proceed to obtain the associated strong problems integrating by parts the terms  $\nabla p_R^i, \nabla p_I^i$ . Noting that  $H_0^1(\Omega) \subset V(\Omega)$  we get

$$\begin{cases} \int_{\Omega} (2u_R^r + \Delta p_R^r + k^2 p_R^r) v_R dx = 0 & \forall v_R \in H_0^1(\Omega) \\ \int_{\Omega} (2u_I^r - \Delta p_I^r + k^2 p_I^r) v_I dx = 0 & \forall v_I \in H_0^1(\Omega) \\ \int_{\Omega} (2u_R^o + \Delta p_R^o + k^2 p_R^o) v_R dx = 0 & \forall v_R \in H_0^1(\Omega) \\ \int_{\Omega} (2u_I^o - \Delta p_I^o + k^2 p_I^o) v_I dx = 0 & \forall v_I \in H_0^1(\Omega), \end{cases} \tag{19}$$

using  $u^i = \tilde{u}^i + \tilde{g}$ . Equation (19) give the strong equation that are valid inside  $\Omega$  as  $H_0^1(\Omega)$  is dense in  $L^2(\Omega)$ . Therefore in particular they are valid for  $v_R, v_I \in V(\Omega)$  and the boundary term of the integration by parts made gives

$$\begin{cases} - \int_{\partial\Omega} \partial_\nu p_R^r \text{Tr } v_R d\mu + \int_{\Gamma_R} \chi (\alpha_I p_I^r - \alpha_R p_R^r) \text{Tr } v_R d\mu = 0 & \forall v_R \in V(\Omega) \\ \int_{\partial\Omega} \partial_\nu p_I^r \text{Tr } v_I d\mu + \int_{\Gamma_R} \chi (\alpha_R p_I^r + \alpha_I p_R^r) \text{Tr } v_I d\mu = 0 & \forall v_I \in V(\Omega) \\ - \int_{\partial\Omega} \partial_\nu p_R^o \text{Tr } v_R d\mu + \int_{\Gamma_R} \chi (\alpha_I p_I^o - \alpha_R p_R^o) \text{Tr } v_R d\mu + \int_{\Gamma_S} k u_I^o \text{Tr } v_R d\mu = 0 & \forall v_R \in V(\Omega) \\ \int_{\partial\Omega} \partial_\nu p_I^o \text{Tr } v_I d\mu + \int_{\Gamma_R} \chi (\alpha_R p_I^o + \alpha_I p_R^o) \text{Tr } v_I d\mu + \int_{\Gamma_S} k u_R^o \text{Tr } v_I d\mu = 0 & \forall v_I \in V(\Omega). \end{cases} \tag{20}$$

From Equation (20) we get directly the equations that  $\partial_\nu p_R^i, \partial_\nu p_I^i$  must satisfy as elements on  $\mathcal{B}'(\partial\Omega)$  considering  $\mathcal{B}(\partial\Omega) = \text{Im}(\text{Tr}|_{V(\Omega)})$ . Since this space is dense in  $L^2(\Gamma_N \cup \Gamma_R, \mu)$ , we get the strong boundary conditions for  $p_R^i, p_I^i$ . Joining this results we get the strong

problems given in Equation (21).

$$\begin{cases} (\Delta + k^2)p_R^r = -2u_R^r & \text{on } \Omega \\ \partial_\nu p_R^r = 0 & \text{on } \Gamma_N \\ p_R^r = 0 & \text{on } \Gamma_D \\ \partial_\nu p_R^r + \chi(\alpha_R p_R^r - \alpha_I p_I^r) = 0 & \text{on } \Gamma_R. \end{cases} \quad \begin{cases} (\Delta + k^2)p_I^r = 2u_I^r & \text{on } \Omega \\ \partial_\nu p_I^r = 0 & \text{on } \Gamma_N \\ p_I^r = 0 & \text{on } \Gamma_D \\ \partial_\nu p_I^r + \chi(\alpha_R p_I^r + \alpha_I p_R^r) = 0 & \text{on } \Gamma_R. \end{cases} \quad (21)$$

$$\begin{cases} (\Delta + k^2)p_R^o = -2u_R^o & \text{on } \Omega \\ p_R^o = 0 & \text{on } \Gamma_D \\ \partial_\nu p_R^o + \chi(\alpha_R p_R^o - \alpha_I p_I^o) = 0 & \text{on } \Gamma_R \\ \partial_\nu p_R^o - k p_I^o = 0 & \text{on } \Gamma_S. \end{cases} \quad \begin{cases} (\Delta + k^2)p_I^o = 2u_I^o & \text{on } \Omega \\ p_I^o = 0 & \text{on } \Gamma_D \\ \partial_\nu p_I^o + \chi(\alpha_R u_I^o + \alpha_I u_R^o) = 0 & \text{on } \Gamma_R \\ \partial_\nu p_I^o + k p_R^o = 0 & \text{on } \Gamma_S. \end{cases}$$

After adding the systems associated to  $p_R^i$  with the systems associated to  $p_I^i$  multiplied by the imaginary unit, the strong adjoint problems are

$$\begin{cases} \Delta p^r + k^2 p^r = -2\bar{u}^r & \text{on } \Omega \\ \partial_\nu p^r = 0 & \text{on } \Gamma_N \\ p^r = 0 & \text{on } \Gamma_D \\ \partial_\nu p^r + \chi \alpha p^r = 0 & \text{on } \Gamma_R, \end{cases} \quad (22)$$

for the reflective case and

$$\begin{cases} \Delta p^o + k^2 p^o = -2\bar{u}^o & \text{on } \Omega \\ p^o = 0 & \text{on } \Gamma_D \\ \partial_\nu p^o + \chi \alpha p^o = 0 & \text{on } \Gamma_R \\ \partial_\nu p^o - i k p^o = 0 & \text{on } \Gamma_S \end{cases} \quad (23)$$

for the open air boundary condition.

Note that the adjoint problems are different, and therefore their solutions are as well. As we did before, we proceed to analyze the terms in the Lagrangians in order to obtain the expressions for  $J'_i$  from the known results. Indeed, the term  $\langle k^2 \tilde{g}, v \rangle_{L^2(\Omega)}$  do not depend on  $\chi$ , so the Fréchet derivative of  $J_1$  is exactly the same as the one obtained on p.115 of the Poly for the Helmholtz problem with homogeneous mixed boundary conditions (no  $\tilde{g}$ ), plus the contribution of the additional term depending on  $\chi$  present on  $FV_1$  :

$$\left\langle \frac{\partial}{\partial \chi} (\chi \alpha \langle \text{Tr } \tilde{g}, \text{Tr } p_r \rangle_{L^2(\Gamma_R, \mu)}), \chi_0 \right\rangle = - \int_{\Gamma_R} \chi_0 \text{Re}(\alpha \text{Tr } \tilde{g} \text{Tr } p_r(\chi)) d\mu. \quad (24)$$

Then

$$\begin{aligned} \langle J'_1(\chi), \chi_0 \rangle &= - \int_{\Gamma_R} \chi_0 \text{Re}(\alpha \text{Tr } \tilde{u}_r(\chi) \text{Tr } p_r(\chi)) d\mu - \int_{\Gamma_R} \chi_0 \text{Re}(\alpha \text{Tr } \tilde{g} \text{Tr } p_r(\chi)) d\mu \\ &= - \int_{\Gamma_R} \chi_0 \text{Re}(\alpha \text{Tr } u_r(\chi) \text{Tr } p_r(\chi)) d\mu \end{aligned} \quad (25)$$

Also, since the terms  $ik \langle \text{Tr } u, \text{Tr } v \rangle_{L^2(\Gamma_S, \mu)}$ ,  $ik \langle \text{Tr } \tilde{g}, \text{Tr } v \rangle_{L^2(\Gamma_S, \mu)}$  do not depend on  $\chi$ , the expression of the Fréchet derivative of  $J_2$  is the same as that of  $J_1$ , but with the values  $p_i, u_i$  replaced accordingly. Then we have

$$\begin{aligned} \langle J'_1(\chi), \chi_0 \rangle &= - \int_{\Gamma_R} \chi_0 \text{Re}(\alpha \text{Tr } u_r(\chi) \text{Tr } p_r(\chi)) d\mu \\ \langle J'_2(\chi), \chi_0 \rangle &= - \int_{\Gamma_R} \chi_0 \text{Re}(\alpha \text{Tr } u_o(\chi) \text{Tr } p_o(\chi)) d\mu. \end{aligned} \quad (26)$$

From this result, we observe that the implementation of both gradients is quite similar and does not require computing  $\tilde{g}$ , as this term is present neither in the adjoint problems nor in the final expression of the derivatives. Its computation is simplified since we already have a code that allows us to solve the Helmholtz problem with non-homogeneous and mixed boundary conditions. Hence, we can compute  $u_r, u_o, p_r, p_o$  by properly modifying the functions `f_dir`, `f_neu`, `f_rob`, and `f_dom`.

The choice of the appropriate modeling of the boundary conditions for our case arises from the physical scenario motivating our work : the absorption of sound waves at the back of a closed concert hall to improve the acoustic resolution of the music played in real time by the orchestra located at the front of the hall. Considering the reverberant characteristics of the surrounding walls, and modeling the hall as a completely closed room with no open windows or doors during the performance, we select fully reflecting walls on both sides of the hall and limit the placement of the absorbent material only to the back wall, which corresponds to the bottom side of our simplified 2D domain  $\Omega$ . This choice is consistent with the fact that, in this scenario, the wave energy cannot escape the room except through dissipation on the absorbent wall that we aim to optimize.

As a second study, we may place absorbent material at strategic points on the sides of the hall while keeping the remaining walls totally reflective, in order to experiment with the optimization methods and better adjust our model [2].

### 3.3 Material choices

Even though audible frequencies range from 20 Hz to 20 kHz, we have chosen to restrict the study to the frequency range of [20 Hz ; 4000 Hz], which corresponds to the frequencies of musical instruments.

To estimate  $\alpha$ , the coefficient that appears in the Robin condition, we use a numerical approach. The key characteristics to identify for each material are porosity, tortuosity, and resistivity. We have also chosen to study the environmental and economic impact of each material to better understand the choices made in practice [5].

| Material   | Porosity $\phi$ | Tortuosity $\alpha_h$ | Resistivity $\sigma$ (kPa · s/m <sup>2</sup> ) |
|------------|-----------------|-----------------------|------------------------------------------------|
| Wool       | 0.985           | 1                     | 1605                                           |
| Polyester  | 0.97            | 1                     | 124000                                         |
| Glass wool | 0.98            | 1.2                   | 20000                                          |
| Isorel     | 0.70            | 1.15                  | 142300                                         |
| Birch LT   | 0.529           | 1.37                  | 151429                                         |
| Melamine   | 0.98            | 1.01                  | 12000                                          |

TABLE 1 – Table of acoustic material properties

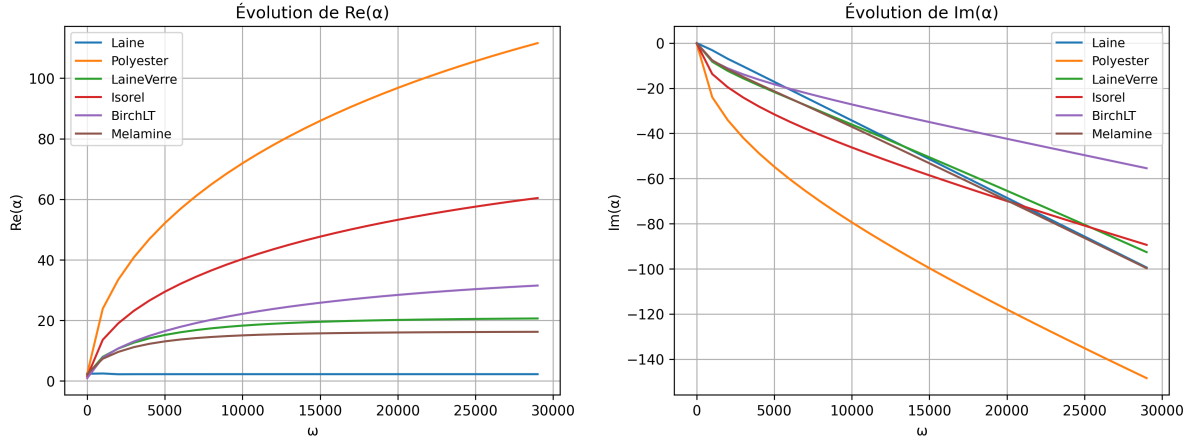


FIGURE 6 – Real and imaginary parts of  $\alpha$  versus angular frequency  $\omega$ , comparing glass wool, isorel, birchlt, melamine, polyester, and wool.

| Material   | Price                           | Environmental impact                                                        | Feasibility / Implementation                                                    |
|------------|---------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Wool       | 25 €/m <sup>2</sup> for 19 mm   | Low embodied energy, renewable material                                     | Easy to handle but mainly used as a decoration                                  |
| Polyester  | 35 €/m <sup>2</sup> for 19 mm   | Recyclable                                                                  | Easy to handle and very used in practice                                        |
| Glass wool | 3 to 25 €/m <sup>2</sup>        | High embodied energy, partially recyclable                                  | Easy to handle and very used in practice despite its environmental limitations. |
| Isorel     | 150 €/m <sup>2</sup> for 19 mm  | Low embodied energy, natural recyclable fiber                               | Easy to handle but not really used to control acoustical energy                 |
| Birch LT   | 43 €/m <sup>2</sup> for 19 mm   | Recyclable and renewable, but requires tree felling (risk of deforestation) | Easy to handle but used for diffusion, not for absorption                       |
| Melamine   | 31.6 €/m <sup>2</sup> for 19 mm | Petrochemical product, non-recyclable                                       | Easy to handle and very used in practice. Good acoustical performances          |

TABLE 2 – Comparison of different materials

We chose to shortlist six materials that may have good acoustical properties. To determine which material we would focus our study on, we used three criteria : price, environmental impact, and feasibility of the solution. We reached the following conclusions :

- Wool can not be used for complex geometrical forms such as fractals.
- Glass wool has a huge environmental impact.
- Isorel is too expensive.

As a result, we decided to study Birch LT and Melamine.

## 4 A first comparison of performances

We performed numerical simulations for the cases  $\beta \in \{0, 1\}$ , which correspond respectively to having no absorbent material on  $\Gamma_R$ , and to covering the entire portion of this boundary with absorbent material, considering a plane wall. The goal is to analyze the behavior of the acoustic energy inside the room under these two extreme conditions, and to identify the critical frequencies that should be optimized.

### 4.1 Preliminary analysis with a reflecting wall

We begin by plotting the total wave energy as a function of the frequency for a perfectly reflecting wall ( $\chi = 0$ ), that is, without any absorbing material. The corresponding result is shown in Figure 7.

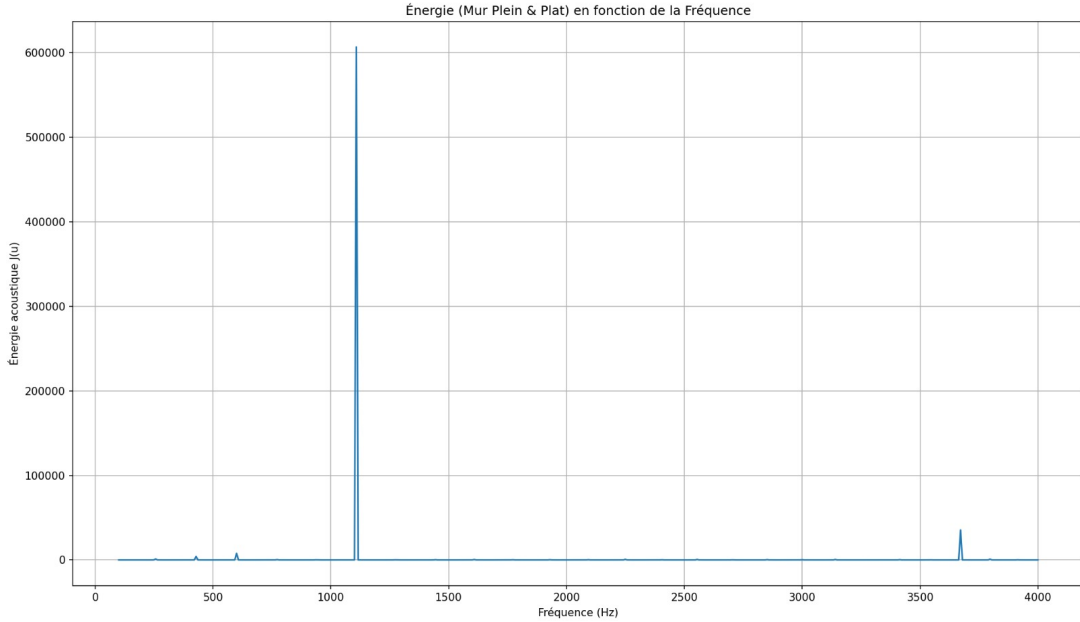


FIGURE 7 – Total wave energy as a function of the frequency for a perfectly reflecting wall ( $\chi = 0$ ).

We observe a strong resonance phenomenon, with a very pronounced energy peak at the frequency  $\omega_0 \approx 1100\text{Hz}$ . However, in practice, no material can be perfectly reflecting ; there is always a non-zero amount of absorption. To account for this, we repeat the calculation with a small absorption coefficient  $\chi = 0.05$ . The resulting plot is shown in Figure 8.

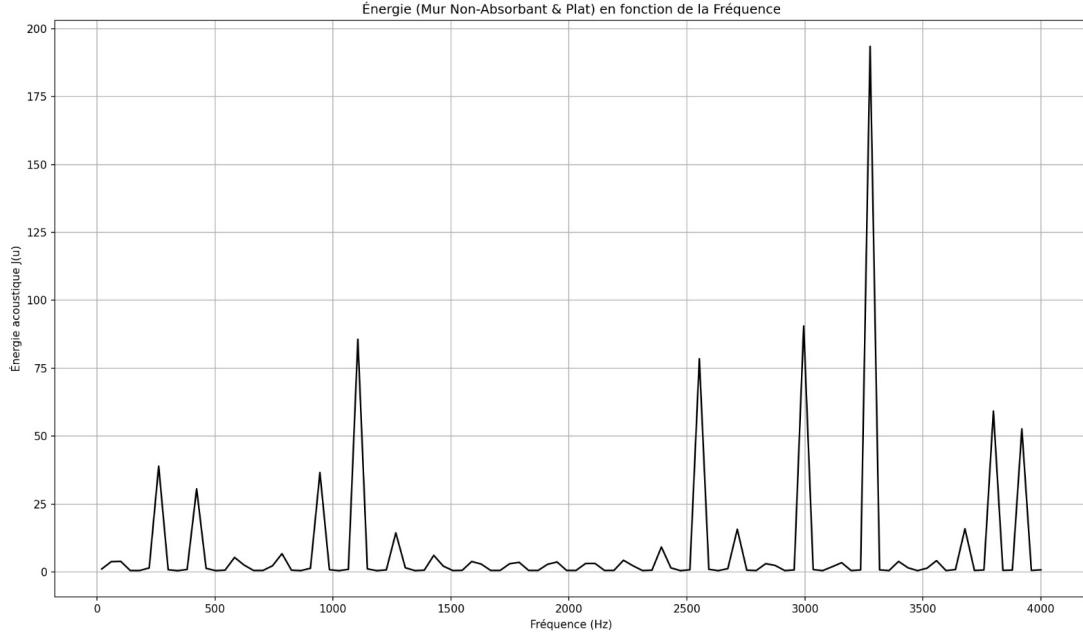


FIGURE 8 – Total wave energy as a function of the frequency for a slightly absorbing wall ( $\chi = 0.05$ ).

Overall, the performance is quite poor for this perfectly reflecting material, with very high energy levels across the whole spectrum.

Then, we compare the behaviour of the wave for different materials by plotting the total wave energy as a function of the frequency. We do this for a fully absorbing wall with a level 0 fractal geometry, and then for a fully absorbing wall with a level 2 fractal geometry. For each configuration, the energy–frequency curve is computed for several materials, which allows us to make a first comparison of performances. We can also assess how the geometric complexity of the boundary influences energy dissipation across the frequency range.

## 4.2 Level 0 fractal – absorbing wall

Figure 9 shows the energy of the wave as a function of the frequency for several materials, in the case of a level 0 fully absorbing wall. The associated geometry used for this experiment is displayed in Figure 10.

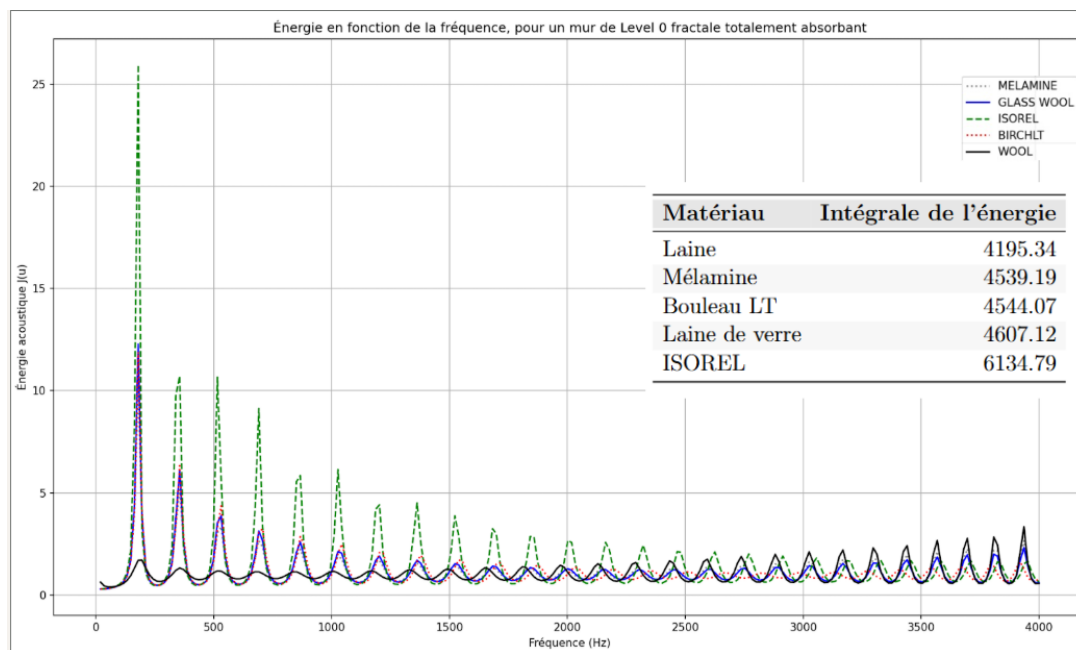


FIGURE 9 – Wave energy as a function of the frequency for a level 0 fully absorbing wall.

FIGURE 10 – Geometry of the level 0 fully absorbing wall.

### 4.3 Level 2 fractal – absorbing wall

We repeat the same analysis for a level 2 fractal wall, still fully absorbing. Figure 11 presents the energy–frequency curve for the same set of materials, while Figure 12 displays the corresponding geometry.

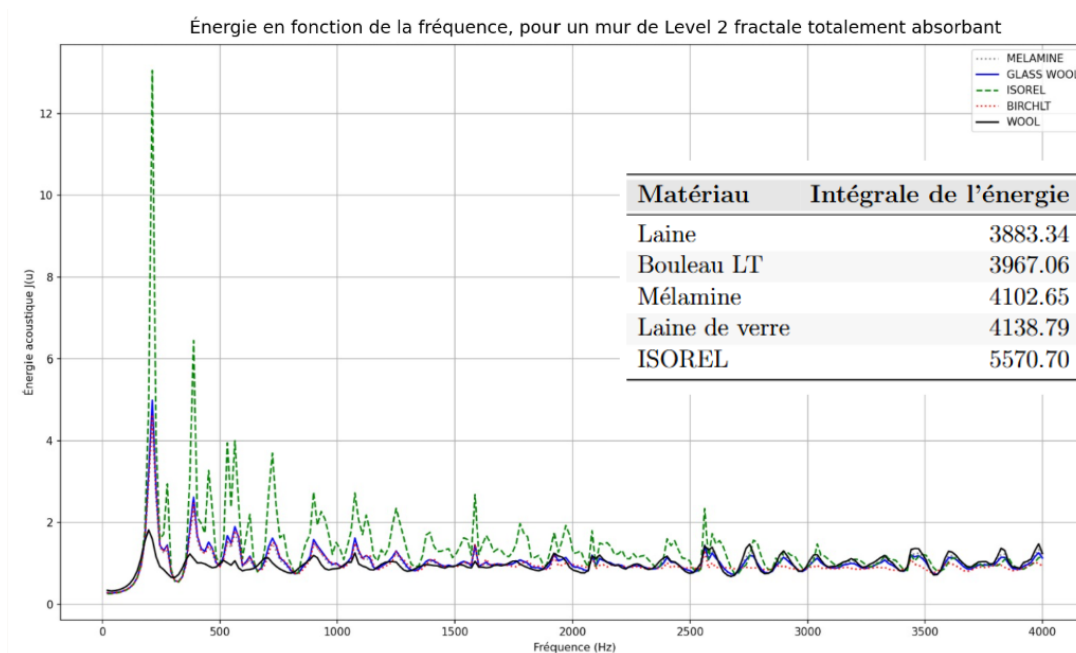


FIGURE 11 – Wave energy as a function of the frequency for a level 2 fully absorbing wall.



FIGURE 12 – Geometry of the level 2 fully absorbing wall.

From these results, we observe that the level-2 fractal geometry provides better performance, with improved energy dissipation compared to the level-0 wall. Among the materials tested, wool yields the best results overall. However, as mentioned earlier, wool cannot realistically be implemented in a fractal geometry.

Moreover, as mentioned earlier, glass wool has a huge environmental impact, and Isorel is too expensive.

For these reasons, we retain melamine and Birch LT as the most suitable candidates : both materials deliver satisfactory performance while remaining compatible with the geometric constraints of the fractal wall.

## 5 Optimization Process Design

We then raise the following questions : can we use less absorbing material while still obtaining equally good, or even better, results? Moreover, for a fixed amount of material, several spatial distributions are possible — which configuration provides the best performance?

Our goal is therefore to determine an optimal quantity of material to place on the wall, and to identify the most effective arrangement of this material to maximise energy dissipation.

The objective of this work is to determine an optimal distribution of absorbing material  $\chi$  along the Robin boundary of the fractal domain in order to minimize the acoustic energy inside the cavity. The design variable  $\chi$  represents a normalized material density, constrained to remain between 0 and 1 and subject to a global volume constraint. The optimization is carried out using a projected gradient descent method relying on the solution of the direct and adjoint Helmholtz problems derived in the previous sections.

### 5.1 Optimization problem

We consider the objective functional

$$J(u, \chi) = \int_{\Omega} |u|^2 dx, \quad (27)$$

where  $u$  is the acoustic pressure field satisfying the Helmholtz equation with Robin boundary damping coefficient  $\alpha(\chi) = \alpha_0 \chi$ . Here  $\alpha_0 \in \mathbb{C}$  is the complex absorption coefficient of the material, and  $\chi$  is extended by zero outside the Robin boundary.

The design variable satisfies the admissibility constraints

$$0 \leq \chi(s) \leq 1 \quad s \in \Gamma_R, \quad (28)$$

as well as a volume (surface) constraint on the total fraction of absorbing material,

$$\frac{1}{|\Gamma_R|} \int_{\Gamma_R} \chi(s) ds = V_{\text{obj}}. \quad (29)$$

The optimization problem therefore reads

$$\min_{\chi \in \mathcal{A}} J(u(\chi), \chi), \quad (30)$$

where  $\mathcal{A}$  is the admissible set defined above.

### 5.2 Sensitivity analysis

Computing the derivative of  $J$  with respect to  $\chi$  requires determining the variation of the state  $u$  with respect to boundary perturbations of the absorption coefficient. Using the adjoint formalism introduced previously, we obtain the pointwise gradient

$$\frac{\partial J}{\partial \chi}(s) = -\Re(\alpha_0 u(s) p(s)), \quad s \in \Gamma_R, \quad (31)$$

where  $p$  is the adjoint state solution of the adjoint Helmholtz problem constructed earlier.

### 5.3 Gradient descent update

Given the gradient  $g = \partial J / \partial \chi$ , a descent step before projection reads

$$\chi^{n+\frac{1}{2}}(s) = \chi^n(s) - \mu_n g(s), \quad (32)$$

where  $\mu_n > 0$  is the current step size. This intermediate update does not yet satisfy the admissibility constraints and must be projected onto  $U_{ab}^*(\beta)$ .

### 5.4 Projection onto the admissible set

The projection consists of enforcing the bounds  $0 \leq \chi \leq 1$  and the volume constraint. First, bound-clipping is performed :

$$\tilde{\chi}(s) = \min\{1, \max\{0, \chi^{n+\frac{1}{2}}(s)\}\}. \quad (33)$$

Then a Lagrange multiplier  $l$  is determined so that

$$\chi^{n+1}(s) = \min\{1, \max\{0, \tilde{\chi}(s) + l\}\} \quad (34)$$

satisfies

$$\frac{1}{|\Gamma_R|} \int_{\Gamma_R} \chi^{n+1}(s) ds = V_{\text{obj}}. \quad (35)$$

Since the left-hand side is a monotone function of  $l$ , the multiplier is found efficiently using dichotomy.

### 5.5 Line search and convergence control

At each iteration, the step size  $\mu_n$  is adjusted with a monotone line-search strategy : if the updated design  $\chi^{n+1}$  decreases the objective energy, the step is slightly increased ; otherwise it is halved until a decrease is observed (or until a minimal threshold is reached). This prevents divergence while accelerating convergence in regions where the functional is locally smooth. But we must understand here that the chosen  $\chi$  takes real values, which is not satisfying in general since for obvious industrial and economical reasons we can't propose a single piece of custom material disposition adapted for a specific case every time a client calls our magnificent Acoufun<sup>TM</sup> company. Thus, in the end, the final  $\chi$  is discretized in  $\{0, 1\}^{\Gamma_R}$  while preserving relation (35). That final  $\chi$  can therefore be properly transposed in real life by placing slabs on the walls.

### 5.6 Resulting algorithm

The overall optimization loop can be summarized as follows :

1. Solve the Helmholtz problem for  $u^n$  with current density  $\chi^n$ .
2. Solve the adjoint problem for  $p^n$ .
3. Compute the gradient  $g^n = -\Re(\alpha_0 u^n p^n)$ .
4. Perform a gradient step  $\chi^{n+\frac{1}{2}} = \chi^n - \mu_n g^n$ .
5. Project  $\chi^{n+\frac{1}{2}}$  onto the admissible set to obtain  $\chi^{n+1}$ .
6. Adjust the step size  $\mu_n$  via a monotone line search.

The iterations are repeated until either the step size falls below a prescribed threshold or the relative variation of the objective functional becomes negligible.

## 6 Performance Comparison for Single-Frequency Optimization

### 6.1 Proposed Distributions of material

We perform several simulations for the materials we selected previously, namely Birch LT and Melamine. For each material, we test multiple percentages of absorbing material and consider all three possible fractal levels of the wall.

For Birch LT, the distribution is optimized for a frequency of 2700 Hz, which corresponds to the frequency where we want to minimize the energy. For Melamine, the optimization is performed at 2550 Hz, which similarly gives the best performance in terms of energy reduction at that frequency. These choices ensure that our optimization targets the frequencies at which the energy is high before the optimization.

Based on these results, we propose the following distributions for the client :

- Melamine covering 50% of the wall, arranged as shown in Figure 13.
- Birch LT covering 60% of the wall, arranged as shown in Figure 14.

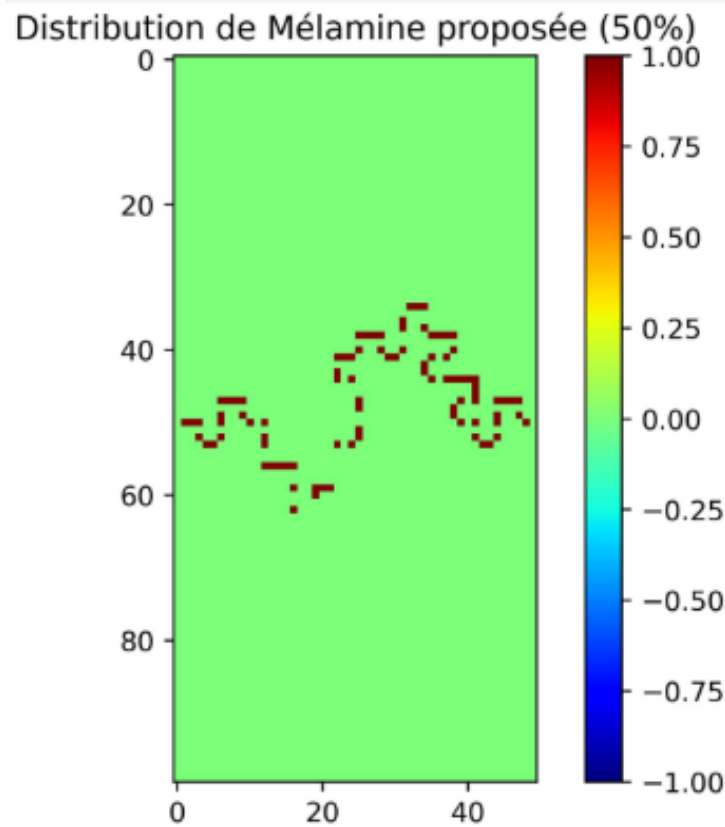


FIGURE 13 – Proposed distribution of Melamine covering 50% of the wall.

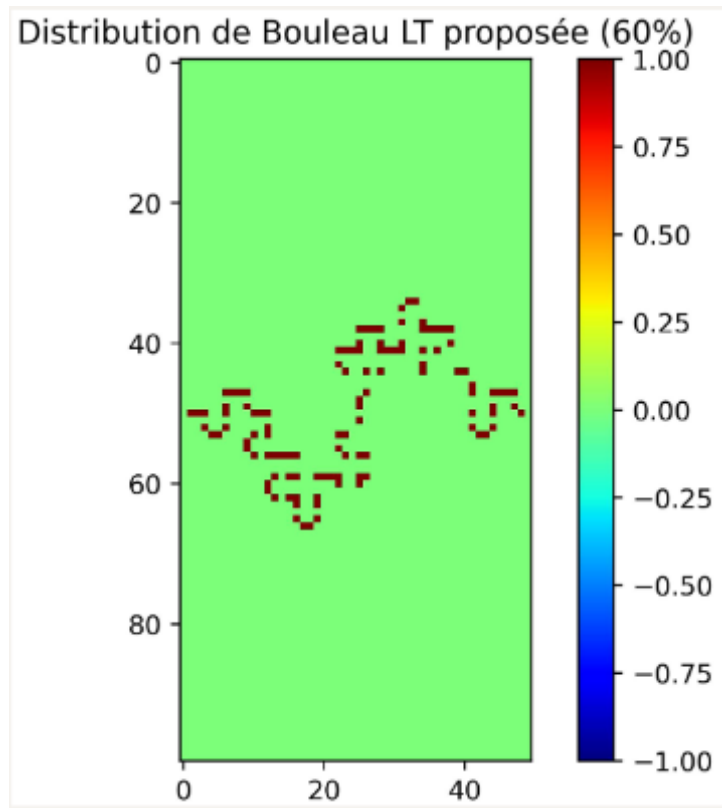


FIGURE 14 – Proposed distribution of Birch LT covering 60% of the wall.

Figures 15 and 16 show the energy curves as a function of frequency for these proposed configurations.

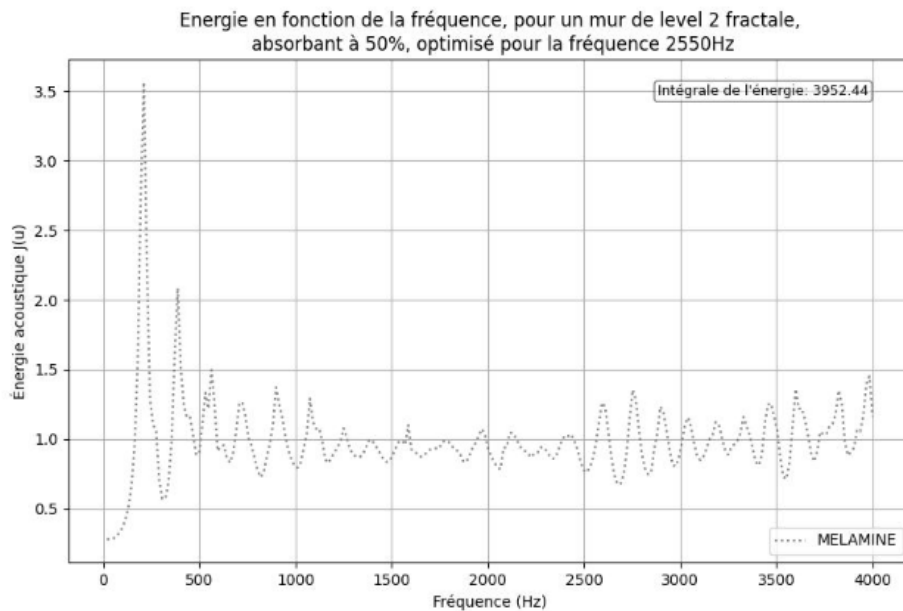


FIGURE 15 – Energy versus frequency for the proposed Melamine distribution.

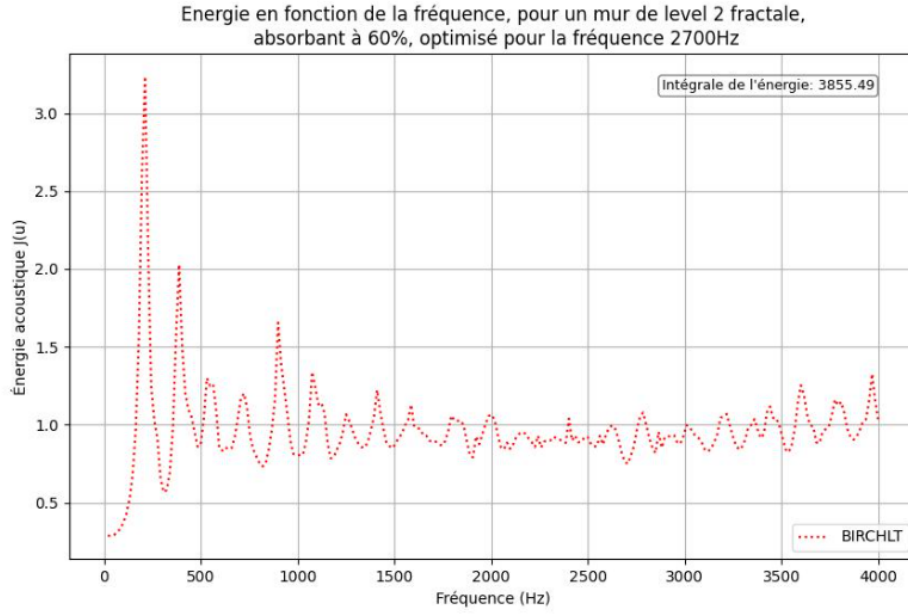


FIGURE 16 – Energy versus frequency for the proposed Birch LT distribution.

During the optimization process, i.e., while the gradient descent algorithm is running, the energy decreases, as expected. This is illustrated in Figure 17.

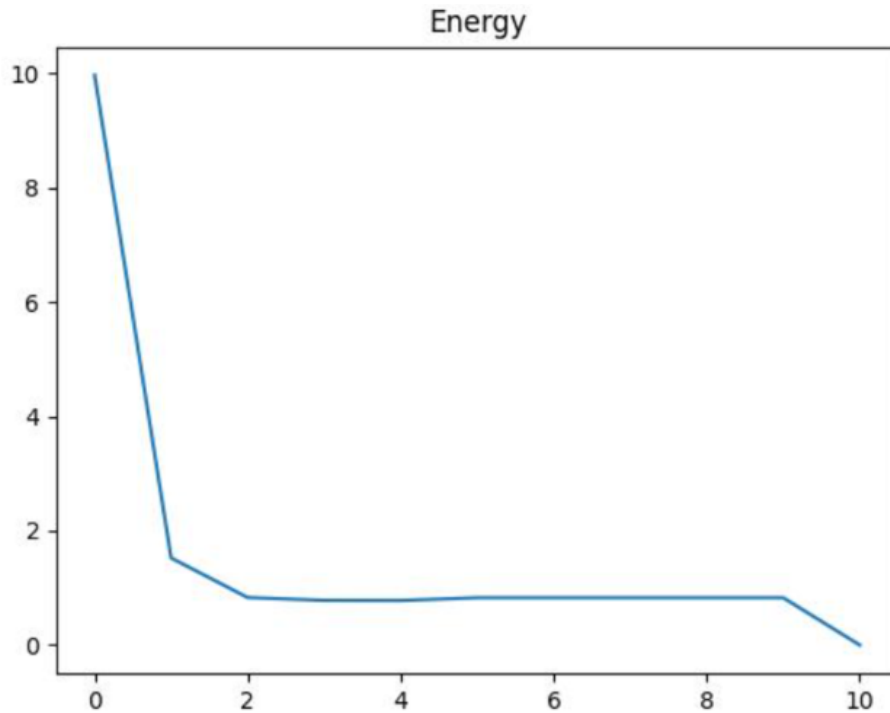


FIGURE 17 – Decrease of the total acoustic energy during the gradient descent optimization.

In both cases, we observe very good results, with low energy across the entire frequency range of interest. Slightly higher energy levels are observed at low frequencies, but this is less problematic for the audience than high-energy peaks at higher frequencies. This explains why we optimized for 2550 Hz and 2700 Hz.

Overall, Birch LT shows slightly better performance, but it is more expensive and requires a larger amount of material. We therefore propose both solutions to the client, who can choose the option that best fits their budget.

To further demonstrate that our proposed distribution is indeed optimal, we can vary the different parameters and compare the resulting energy curves. For instance, we plot the energy as a function of frequency using only 40% of Melamine (instead of the 50% used in our proposed solution). As shown in Figure 18, the energy is higher than with our optimized configuration (the curves may look similar but the integral of the energy is lower with our optimized configuration).

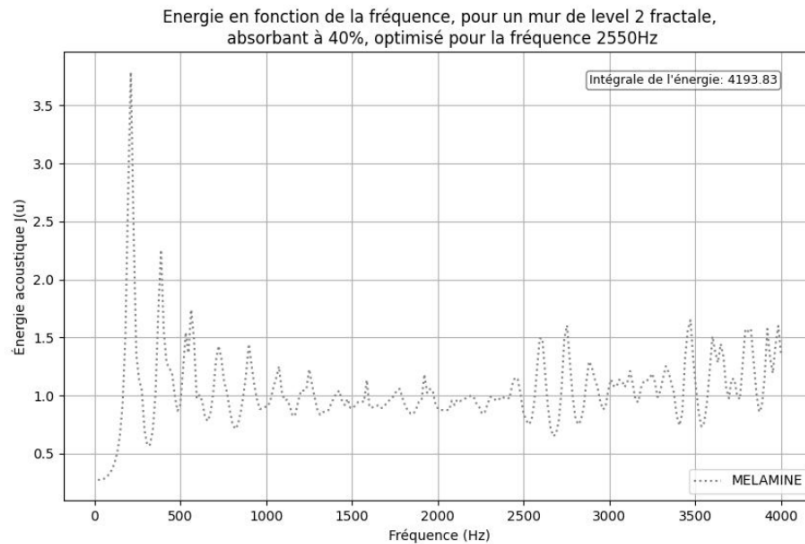


FIGURE 18 – Energy versus frequency using 40% of Melamine. The energy is higher than in our optimized solution.

We can also test the same amount of material on a level 1 fractal wall. The resulting curve, shown in Figure 19, again exhibits higher energy levels, confirming that both the percentage of material and the fractal geometry used in our solution are close to optimal.

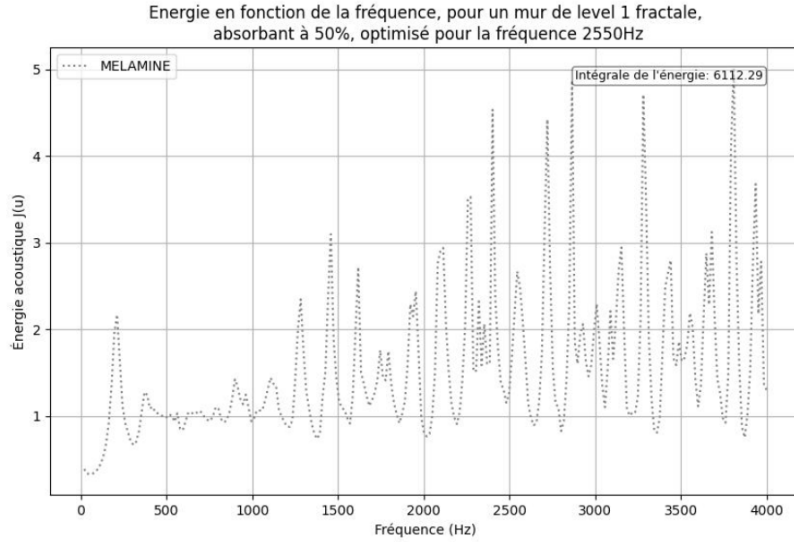


FIGURE 19 – Energy versus frequency for 50% Melamine on a level 1 fractal wall. The performance is worse than with our proposed geometry.

To obtain a more complete and systematic understanding of the behaviour of each material, we compile the results into summary tables. For both Birch LT and Melamine, we list the total acoustic energy obtained for all tested percentages of absorbing material, and for fractal levels 1 and 2 (level 0 is excluded because it does not provide satisfactory performance). These tables allow us to clearly compare how the material quantity and the fractal geometry influence the global energy performance.

**Birch LT — Total Acoustic Energy (kJ)**

| $\beta$ Fractal | 10%   | 20%   | 30%  | 40%  | 50%  | 60%  | 70%  | 80%  | 90%  |
|-----------------|-------|-------|------|------|------|------|------|------|------|
| Level 1         | 57.84 | 33.05 | 9.58 | 8.89 | 5.21 | 4.06 | 4.06 | 4.04 | 4.05 |
| Level 2         | 17.62 | 8.72  | 4.03 | 3.98 | 3.88 | 3.86 | 3.86 | 3.87 | 3.87 |

TABLE 3 – Energy (in kJ) of Birch LT with fractal wall level 1 or 2 for different absorbent percentages

**Melamine — Total Acoustic Energy (kJ)**

| $\beta$ Fractal | 10%   | 20%   | 30%   | 40%  | 50%  | 60%  | 70%  | 80%  | 90%  |
|-----------------|-------|-------|-------|------|------|------|------|------|------|
| Level 1         | 38.64 | 24.86 | 16.63 | 6.11 | 4.17 | 4.35 | 4.35 | 4.36 | 4.36 |
| Level 2         | 27.81 | 12.72 | 6.17  | 4.19 | 3.95 | 4.04 | 4.05 | 4.06 | 4.05 |

TABLE 4 – Energy (in kJ) of Melamine with fractal wall level 1 or 2 for different absorbent percentages



## 6.2 Performance analysis of our solutions on the real source (the orchestra)

We now perform a performance analysis under conditions that are closer to the real situation during a concert. In what follows, we analyse the behaviour of our proposed solution for Birch LT. (The results for the Melamine configuration are similar and lead to the same conclusions.)

### La 440 Hz

We first evaluate the performances for a La 440 Hz note played by the orchestra. As described in Section 3.1, only the left and central sections of the orchestra are able to produce this note during a performance. We therefore model the excitation by two sources : one on the left and one in the centre, each emitting a spherical wave at 440 Hz.

Figure 20 shows the real part of the resulting acoustic field for the three configurations :

- a reflecting wall (no absorbing material),
- a wall composed of 60% Birch LT with the *initial, non-optimized* distribution,
- our proposed solution (60% Birch LT with the optimized distribution).

The corresponding energy values are also displayed for comparison.

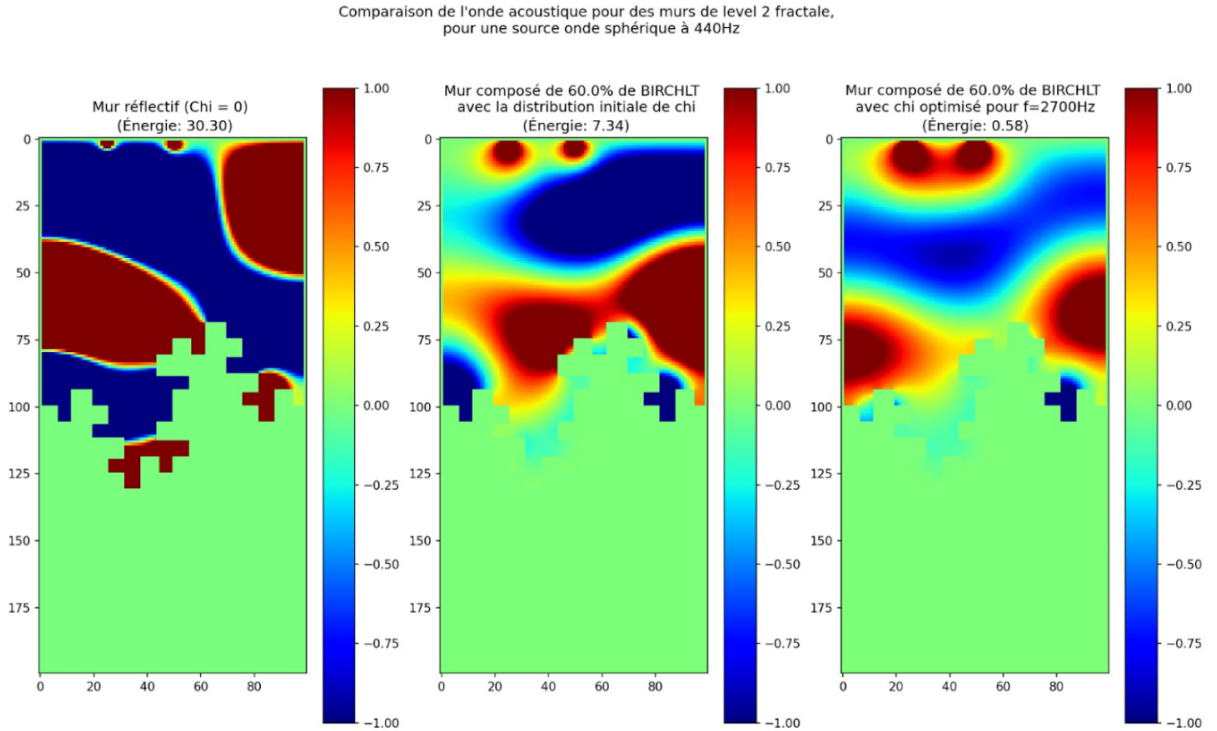


FIGURE 20 – Real part of the acoustic wave at 440 Hz for the three configurations : reflecting wall, non-optimized Birch LT distribution, and optimized Birch LT solution.

On this plot, one can clearly identify the two acoustic sources corresponding to the left and central parts of the orchestra playing the La 440 Hz. As expected, the reflecting wall provides the worst performance, with very high energy levels. We also observe that our

optimized solution significantly reduces the energy compared to the initial, non-optimized distribution.

This demonstrates that our design performs well even though the optimization was carried out at 2700 Hz : it still provides robust and efficient damping at lower frequencies.

### Other relevant notes

We repeat the same analysis for several musically relevant notes likely to be produced during a concert. For instance, for a Do 1047 Hz (which can be played by the left, central, and right sections of the orchestra), the results are shown in Figure 21.

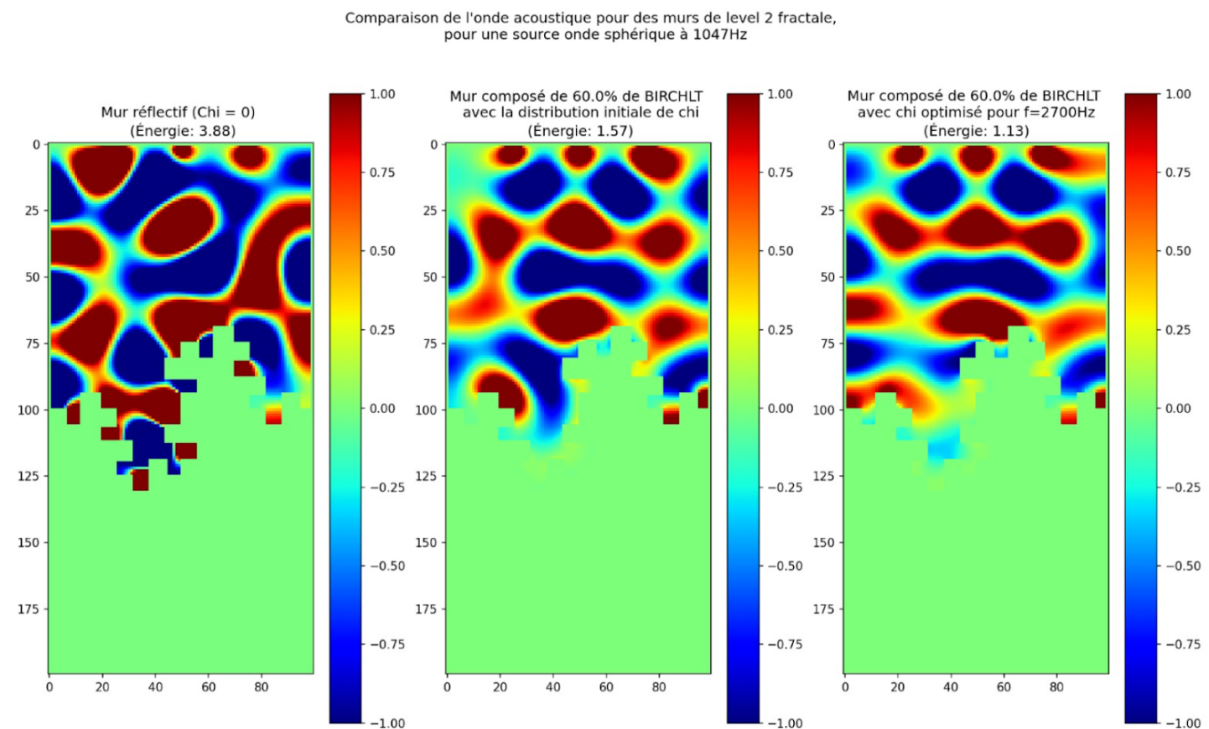


FIGURE 21 – Real part of the acoustic wave at 1047 Hz for the three configurations : reflecting wall, non-optimized Birch LT distribution, and optimized Birch LT solution.

Again, this proves that our solution provides good results.

Thus, with our first method, which relies on single-frequency optimization, *AcouFun* recommends using BirchLT at 60% or Melamine at 50%, arranged as shown in Figures 13 and 14.

However, our startup is constantly seeking to improve and refine its methods. As a perspective, we envisage performing an optimization over multiple frequencies rather than a single one. This is the objective of the next section.

## 7 Multi-Frequency Optimization

Real acoustic fields produced by musical instruments or loudspeaker systems cannot be adequately represented by a single monochromatic excitation. In the present setting, three distinct localized sources are considered, each corresponding to a different family of instruments. Their spectral content is encoded through frequency-dependent boundary load functions  $g_j(f)$ , where  $j = 1, 2, 3$  indexes the instrument types and  $f \in \mathcal{F}$  denotes the discrete set of frequencies composing the excitation spectrum.

For a given design  $\chi$  of the absorbing material distribution, solving the Helmholtz problem at multiple frequencies yields a set of pressure fields  $\{u_{j,f}\}$ , one per source and per frequency. The acoustic energy within the cavity is therefore frequency-dependent, and the objective functional must capture contributions from all frequencies and all sources.

### 7.1 Multi-frequency objective functional

Let  $\mathcal{F} = \{f_1, \dots, f_{N_f}\}$  be the sampled frequency set, and let  $w(f) \geq 0$  be a spectral weight representing the amplitude of the excitation at frequency  $f$ . The total objective functional is defined as the weighted sum

$$J(\chi) = \sum_{j=1}^3 \sum_{f \in \mathcal{F}} w(f) J_{j,f}(\chi), \quad (36)$$

where

$$J_{j,f}(\chi) = \int_{\Omega} |u_{j,f}|^2 dx \quad (37)$$

is the acoustic energy associated with instrument  $j$  at frequency  $f$ .

Each field  $u_{j,f}$  solves the Helmholtz problem at frequency  $f$  with boundary excitation  $g_j(f)$  supported only on the localized portion of the boundary associated with instrument  $j$ .

### 7.2 Adjoint problems and frequency-wise sensitivities

Following the same variational principles as in the monochromatic case, each functional  $J_{j,f}$  admits an adjoint field  $p_{j,f}$  solving the adjoint Helmholtz problem at the same frequency  $f$ .

The sensitivity of  $J_{j,f}$  with respect to a variation  $\delta\chi$  of the design variable is

$$dJ_f(\chi)[\delta\chi] = -\Re\left(\int_{\Gamma_R} \alpha_0 u_f p_f \delta\chi ds\right). \quad (38)$$

Hence, the pointwise gradient contribution from a single frequency is

$$g_f(s) = \frac{\partial J_f}{\partial \chi}(s) = -\Re(\alpha_0 u_f(s) p_f(s)). \quad (39)$$

### 7.3 Weighted sum of gradients

Since the global objective is a weighted sum of frequency-dependent energies, its gradient is the corresponding weighted sum of frequency-wise gradients :

$$g(s) = \frac{\partial J}{\partial \chi}(s) = \sum_{f \in \mathcal{F}} w(f) g_f(s). \quad (40)$$

This aggregated gradient combines the contributions of :

- the spectral content of each instrument (through  $w(f)$ ),
- the frequency-dependent propagation and absorption phenomena (since both  $u_f$  and  $p_f$  depend on  $f$ ).

The use of a weighted sum ensures that the optimizer reduces energy in the frequency bands where the excitation is strongest while still accounting for secondary harmonics and multi-source interactions.

### 7.4 Gradient descent and projection

The descent step remains structurally identical to the single-frequency case :

$$\chi^{n+\frac{1}{2}}(s) = \chi^n(s) - \mu_n g(s), \quad (41)$$

followed by projection onto the admissible set  $0 \leq \chi \leq 1$  and enforcement of the volume constraint using a Lagrange multiplier.

The only modification lies in the computation of the gradient : instead of solving one forward/adjoint pair, one must solve a set of  $|\mathcal{F}|$  forward problems and the same number of adjoint problems. The aggregated gradient is then formed by summing all contributions.

### 7.5 Algorithmic structure

The multi-frequency optimization loop is therefore :

1. For each each frequency  $f \in \mathcal{F}$  :
  - (a) Solve the forward Helmholtz problem for  $u_f^n$ .
  - (b) Solve the adjoint problem for  $p_f^n$ .
  - (c) Compute the partial gradient  $g_f^n = -\Re(\alpha_0 u_f^n p_f^n)$ .
2. Form the weighted gradient

$$g^n = \sum_{f \in \mathcal{F}} w(f) g_f^n.$$

3. Perform a gradient step  $\chi^{n+\frac{1}{2}} = \chi^n - \mu_n g^n$ .
4. Project onto the admissible set to obtain  $\chi^{n+1}$ .
5. Update the step size through monotone line search.

This multi-frequency framework naturally promotes absorbing material placement that is robust across a spectrum of excitations and across multiple instruments localized at different positions on the source boundary.

## 7.6 Source construction and spatial discretization

The multi-frequency source term used throughout the optimization is constructed from three localized boundary regions, each associated with a different instrument family. For each family, we prescribe a *main frequency* and generate a small set of neighboring frequencies to emulate realistic spectral content. In the numerical implementation, we restrict to one dominant frequency per band (i.e.,  $n_{\text{harm}} = 0$ ) in order to isolate the response of the cavity to the primary tones. The corresponding load functions  $g_j(f)$  are obtained from the routine `create_three_band_harmonic_sources`, which assigns each frequency to a spatially separated boundary patch. This separation ensures that the three instrument families interact only through the propagated acoustic field inside the cavity.

The spatial mesh is defined by  $M = 2N$  points in the horizontal direction and  $N$  points in the vertical direction. The grid spacing is chosen as

$$\Delta x = \Delta y = \frac{1}{4N},$$

which provides twenty grid cells per unit length at the scale  $1/N$  used to resolve the cavity geometry. This value was found to yield a good compromise between numerical accuracy and computational cost, especially at the highest frequencies considered (around 2.5 kHz), for which proper sampling of the acoustic wavelength is essential.

## 7.7 Optimization repeated for each material

The multi-frequency optimization procedure described above is executed *independently for each candidate absorbing material*. Each material type enters the Helmholtz model through the frequency-dependent Robin coefficient  $\alpha(f, \text{material})$ , which encodes the acoustic impedance properties of the material. For a fixed material, the optimization produces :

- the optimal density distribution  $\chi_{\text{opt}}$ ,
- the corresponding acoustic field  $u_{\text{opt},f}$  for each frequency,
- and the minimized multi-frequency energy  $J_{\text{opt}}$ .

This ensures that materials are compared fairly : they are all subjected to the *same geometry, same sources, same discretization, and same optimization procedure*. Any difference in performance reflects purely their acoustic absorption behavior.

## 7.8 Selection of the best material

To determine which material yields the most effective absorption, we compute, for each material  $m$ , its minimized objective value

$$J_{\text{opt}}^{(m)} = \sum_{f \in \mathcal{F}} w(f) J_f(\chi_{\text{opt}}^{(m)}),$$

during the optimization. The material achieving the smallest final energy

$$m^* = \arg \min_m J_{\text{opt}}^{(m)}$$

is then selected as the most effective for multi-frequency acoustic damping.

The energy distributions across frequencies for all tested materials are shown in Figure 22. This plot allows direct visual comparison of how each material performs over the entire frequency range, revealing in particular which materials absorb low, medium, or high frequencies most efficiently.

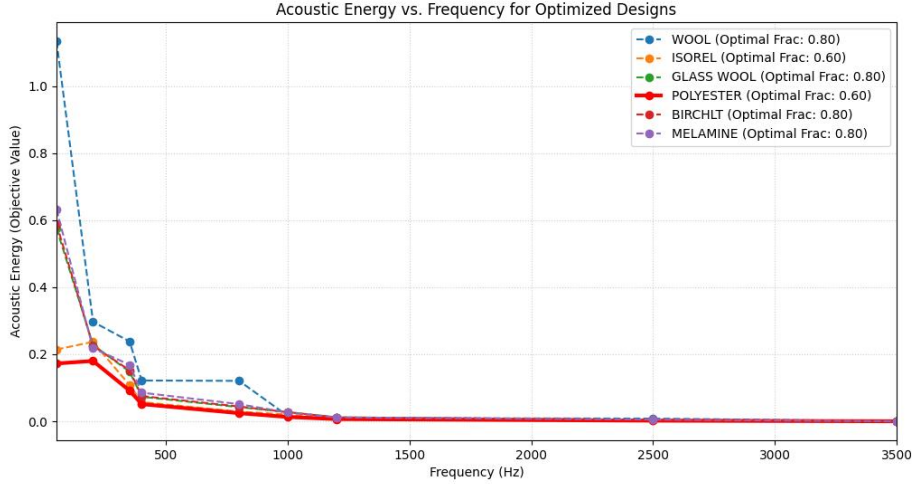


FIGURE 22 – Final minimized energy  $J_f(\chi_{\text{opt}})$  for each frequency and each tested material. This sweep illustrates the frequency-dependent performance of the materials and is used to identify the globally optimal material.

## 7.9 Interpretation of the Optimization Results

The frequency sweep presented in Figure 22 allows direct comparison of the energy-minimizing performance of each tested absorbing material. Among all candidates, the polyester material with parameter  $\beta_{\text{pol}} = 0.60$  achieves the lowest overall multi-frequency energy. This indicates that, when optimized independently in the multi-frequency framework, the polyester provides the most effective damping across the combined set of source bands.

However, the material *isorel* with parameter  $\beta_{\text{iso}} = 0.80$  yields an energy profile that is consistently close to the optimal one. More importantly, a frequency-by-frequency inspection reveals that *isorel* performs better at several individual frequencies, particularly in the mid-frequency range. However we don't select it because it is too expensive, as mentioned earlier

BirchLT also is very efficient, which agrees with the trends already observed in the single-frequency optimization experiments, where BirchLT emerged as one of the most coherent high-performance solutions.

For this reason, although the polyester at  $\beta = 0.60$  is nominally the global minimizer of the multi-frequency objective, the BirchLT material at  $\beta = 0.80$  provides a more *globally coherent* behavior across all frequency regimes and across both single-frequency and multi-frequency optimizations. In practical acoustic design, such consistency is often more important than small numerical differences in the global objective value.

We therefore select BirchLT with  $\beta = 0.80$  as the recommended material for the cavity configuration studied here, as it offers the best compromise between robust multi-band

absorption and stability of performance across all tested optimization scenarios.

## 8 Conclusion - AcouFun Solution

Using our first optimization method, AcouFun recommends to the client either BirchLT at 60% coverage or Melamine at 50%, placed as shown in Figures 13 and 14. The BirchLT solution provides slightly better performance in terms of acoustic energy reduction, but it requires a higher quantity of material and is more expensive than Melamine. Therefore, the final choice depends on the client's budget and preferences.

Looking forward with our proposed second method, which involves multi-frequency optimization, AcouFun shows that Polyester also represents a solid choice. BirchLT remains highly effective under this broader approach, reinforcing the idea that it is a reliable material for acoustic optimization.

In conclusion, our study demonstrates the importance of both material selection and optimal spatial distribution of absorbing surfaces. The fractal geometry of the walls improves energy dissipation, and careful optimization ensures that even a limited amount of material can provide very good acoustic performance across the frequencies of interest.

Furthermore, the methodology developed here is adaptable to other concert conditions, considering the specific positions of instrument sections and relevant frequencies.

Finally, these results highlight the potential for AcouFun to offer tailored acoustic solutions that balance performance, material costs, and feasibility, while opening the door to further improvements through multi-frequency and multi-source optimization.

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